

GuidePath technical memo

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Tactile paving inspection using RT-DETR and structured reasoning

GuidePath is a prototype for reviewing photographs of tactile paving and nearby bicycles or motorcycles. I selected this problem because useful inspection requires both object detection and a defensible interpretation of relative position. The intended user is a maintenance reviewer who decides whether a location needs inspection.

Current scope: AI-assisted implementation and dataset investigation, with an initial training smoke check. Final model performance and deployment remain unverified. This memo presents the evidence available and the decisions behind the work.

Problem definition and practical value

The system estimates possible obstruction in the visible image. It does not certify walking clearance or route safety. A motorcycle beside a path and one overlapping it should produce different evidence. The custom **tactile_paving** class meets the non-COCO requirement; bicycle and motorcycle provide a deliberately limited obstacle vocabulary. Boxes, poles and other hazards remain outside this version.

Dataset selection and evidence

The initial candidate contained paving labels without the required obstacle coverage. I moved to WOTR, whose author repository describes VOC annotations for tactile paving and street objects. The converter maps **blind_road** to **tactile_paving**, retains bicycle and motorcycle, and keeps background images. The initial audit produced the following counts, before cleanup.

Split	Images	Paving instances	Bicycles	Motorcycles
Train	9,056	1,596	4,012	8,059
Validation	2,338	371	987	1,887
Test	2,534	409	992	2,202

Finding: the audit raised **34 flags**, including invalid selected-class boxes and exact duplicates across splits. For example, training image 10007690 matched validation image 30007264; training image 10007030 matched test image 20006850. These are evidence of leakage risk, not 34 distinct bad images.

Response: the cleanup procedure excludes whole images with invalid selected-class boxes and retains one exact copy, preferring test, then validation, then training. Original outputs and exclusion reasons are preserved. Final cleaned counts are not asserted here. Exact deduplication does not resolve near-duplicate or location leakage; those require additional review.

Source: Xia et al., A dataset for the visually impaired walk on the road, Displays 79 (2023), 102486. Author repository: github.com/kxzt/WOTR. Counts and flags above are from the project audit shared during development.

API design and reasoning boundaries

POST /detect accepts an image and returns classes, pixel bounding boxes and detection confidence. **POST /reason** accepts an image and question. Handwritten routing handles counts, visible objects and possible obstruction; capability questions skip detection and unsupported questions receive an explicit limitation.

For each supported obstacle, code calculates the intersection with a paving box divided by the obstacle-box area. Candidate overlap and confidence thresholds are provisional until validation. The response links obstacle and paving IDs to the evidence. Detection confidence is never presented as a probability of physical blockage.

The three assessment states are **possible_obstruction**, **no_obstruction_detected** and **insufficient_information**. In a software test with a bicycle detection but no confident paving detection, the rule returns: "Insufficient information: tactile paving was not detected confidently." This is a test fixture, not a claimed field observation.

The direct LLM adapter selects a typed plan: a conclusion, supporting fact IDs and caveats. Code checks the decision and IDs, then renders verified facts into plain language. A live provider call remains unverified. No agentic framework is used. Missing trained weights produce an explicit error; illustrative demo inputs are kept separate.

Evaluation and five failure investigations

The evaluation scripts target per-class mAP at IoU 0.50 and 0.50 to 0.95, precision, recall and confusion behaviour. These measure detection on the selected split, not route accessibility. The planned reasoning evaluation compares overlap rules with a co-occurrence baseline and reports false alarms and abstentions. Thresholds must be selected on validation data before final testing.

Five model failure cases are not yet verified. The following investigations are planned and must be replaced by actual image IDs, predictions and supported root-cause analysis to meet RAP's failure-case requirement. Proposed fixes are untested.

Investigation	Evidence to inspect and proposed response
Diagonal paving	An object beside the tiles may overlap the rectangular box. Inspect geometry; consider paving-region masks in a later version.
Small or distant paving	Compare paving misses with target size. Review label coverage and test resolution changes on validation images.
Partial occlusion	Inspect hidden paving and weak detections. Assess whether abstention correctly prevents a clear-path claim.
Shadows or poor light	Compare original images, labels and predictions. Test targeted augmentation only if failures support it.
Bicycle and motorcycle confusion	Inspect class swaps and count errors. Review ambiguous labels and collect representative hard examples.

Reproducibility and delivery status

The package contains training and evaluation scripts, a Colab notebook, FastAPI code, tests and a Dockerfile. **29 development software tests passed**; Colab showed a Tesla T4 and a successful one-epoch smoke command. Neither proves model quality. Training scripts record runtime settings, elapsed time, environment versions and a dataset-manifest hash.

Outstanding: a verified deliverable checkpoint, final training record and metrics, five observed failures, live LLM verification, reachable deployment and a public source repository. The memo supports review of the approach; it does not replace those required deliverables. The project was developed with AI assistance, and its design choices and code require my verbal defence.